

Mikhail Razakov

Quantitative Developer / Research Software

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My background is compiler/program-analysis engineering: C++/Python, algorithmic problem solving, reproducible test harnesses, and low-level code. For Quantitative Developer / Research Software roles, the most relevant projects are a PS-form analyzer with an independent exact oracle and an x86-64 backend with differential execution.

EXPERIENCE

July — August
2026

MCST — Compiler Engineering Intern

- Implemented an LLVM 22 LICM pass with loop analysis, side-effect and speculative-safety checks, and hoisting of loop-invariant instructions to a preheader.
- Developed a conservative interprocedural global-IV prototype with APInt affine evolution, transitive call effects, a separate legality phase, and an LLVM IR transform; unsupported CFG and call cases are left unchanged.

PROJECTS

PS-form Memory Dependence Analyzer

Conservative analysis of parametric memory-access overlap using normalization, interval and modular/GCD constraints, exact affine cases, and guarded fallbacks.

Unproved cases return maybe. Small domains are checked by an independent exact oracle, with randomized/metamorphic generation and reproducible counterexamples.

x86-64 Codegen & Register Allocation Lab

An educational compiler backend with SSA/CFG validation, liveness/interference, register allocation, phi lowering, and an x86-64 emitter.

Assignments are checked by a separate verifier; native code runs in a child process and is differentially compared with the IR interpreter.

UniversalToolchain

Modular infrastructure for language/compiler pipelines: dependencies, providers, pass order, and artifact routes are resolved before execution and captured in a LanguagePlan.

Runtime executes the selected plan without a second resolver. The canonical test contract currently records 1,306 passing tests, with separate binding and interpreter/CIL parity checks.

SKILLS

Programming

C++23, Python, C17, C#, Rust

Algorithms & analysis

graph algorithms, data structures, program analysis, CFG/SSA, compiler optimizations

Experimental tooling

exact small-domain oracles, differential/metamorphic/randomized tests, reproducible Python harnesses

Low-level

LLVM, x86-64, code generation, register allocation, Linux, CMake

EDUCATION

HSE University — Software Engineering, 2026–2030

RECOGNITION

UniversalToolchain — Baltic Science and Engineering Competition

First-degree diploma and the Grand Prize "Perfection as Hope" for UniversalToolchain.

MEPhI Junior

First-degree diploma, 96/100 for the UniversalToolchain project.

Moscow / remote